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Mnemonic emotion regulation: a three-process model

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ABSTRACT

Emotion regulation comprises attempts to influence when and how emotions are experienced and expressed. It has mostly been conceived of as proactive (e.g. situation selection) or reactive (e.g. attentional distraction), but it may also be retroactive and involve memory. I term such past-oriented activity mnemonic emotion regulation and propose that it involves increasing or decreasing access to or altering the characteristics of a memory. People may increase access to a memory and make it more likely that it will be retrieved in the future, for example by rehearsing a pleasant memory. They may decrease access to a memory and make it less likely that intrusions will be experienced in the future, for example by repeatedly trying to stop an unpleasant memory from being retrieved. Finally, people may alter the characteristics of a memory and change its content or context, for example by replacing a negative impression with a productive interpretation. I discuss how mnemonic emotion regulation may be instigated (e.g. via elaborate rehearsal) as well as the different motives (e.g. hedonic motives) people may have for engaging in regulation. Also, I discuss possible benefits of, variations in, and improvements of mnemonic emotion regulation.

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Memories often evoke emotions (see Holland & Kensinger, 2010). For example, remembering having been applauded for completing a successful presentation may elicit joy and pride. Such a recollection may make an individual feel good in the moment; it may give rise to pleasant interoceptive sensations, encouraging thoughts as well as a desire to re-experience similar situations in the future. However, memories do not only evoke emotions on a moment-to-moment basis. The balance of positive compared to negative recollections over a longer time period may influence how an individual feels in general. The emotional tone of prevalent and salient memories may colour the view of the self and make the world and the future seem bright and promising or dark and hopeless. Positive autobiographical memories can facilitate a productive view of the self in mentally healthy people (Conway, Singer, & Tagini, 2004). Also, a tendency to recall many positive and few negative memories in unconstrained settings has been associated with high ratings of subjective well-being (Otake, 2015). Conversely, having difficulties

remembering positive experiences and recalling an excessive amount of negative memories has been associated with affective disorders such as anxiety and depression (Dalgleish & Werner-Seidler, 2014; Mitte, 2008). Accordingly, sometimes memories are treasurable and evoke desired emotions, but at other times they are burdensome and spur disliked emotions. If it is possible to control which memories mostly come to mind, then it may influence emotional experiences as well as the ways the self and the world are assessed. Specifically, it may be possible and beneficial to command memories to conduct emotion regulation.

Emotion regulation

Emotion regulation comprises attempts to influence when and how different emotions are experienced and expressed and it involves changing the duration, intensity and/or nature of emotions (Gross, 1998, 2015; Koole, 2009). It is a process that concerns how people respond to and manage emotion-generating situations, so they are not reflexively carried away. Emotion

regulation is typically motivated by hedonic goals (Gross, Richards, & John, 2006); people mostly try to downregulate negative emotions such as anger and sadness and upregulate positive emotions such as joy and interest, possibly because they are inherently motivated to minimise pain and maximise pleasure (see Higgins, 2014). For example, an individual who is bothered by angry thoughts about an unpleasant disagreement at work may try to distract him- or herself by focusing on a promising assignment in order to feel better. However, people occasionally engage in counter-hedonic emotion regulation based on instrumental motives. For instance, an individual may attempt to increase anger with the expectation that it will help succeeding in a negotiation (Tamir & Ford, 2012). Emotion regulation is often deliberate, but it may sometimes become automatic (Bargh & Williams, 2007). For instance, an individual may learn to automatically use distraction to quell anger when in presence of a superior at work. Moreover, emotion regulation affects several important outcomes, such as social interaction (e.g. avoiding becoming angry in delicate settings; see English, Lee, John, & Gross, 2017) and mental health (e.g. dampening irrational fear; see Sheppes, Suri, & Gross, 2015).

Emotion regulation may be understood in different ways (Koole, 2009), but one prominent theoretical conceptualisation is the process model of emotion regulation (e.g. Gross, 1998, 2015; Gross et al., 2006). According to this model, regulation may affect diverse behavioural and mental phenomena at different stages in the emotion-generative process; it may involve situation selection (e.g. avoiding a potentially unpleasant situation), situation modification (e.g. changing the dynamics of an unpleasant situation), attentional deployment (e.g. using distraction to shift attention away from a negative stimulus), cognitive change (e.g. reframing a negative stimulus) and response modulation (e.g. suppressing an undesirable motor response). This conceptualisation highlights proactive (future-oriented) and reactive (present-oriented) but not retroactive (past-oriented) emotion regulation involving memory. Specifically, the model focusses on dynamically unfolding emotional and regulative processes, but it does not include a stage comprising emotion regulation that involves memory or otherwise specify how memory may be a basis for or a target of emotion regulation. This is evidenced by authoritative expositions of the process model that make almost no mentioning of memory (Gross, 2015; Gross et al., 2006; Gross & Thompson, 2007). This inattention to the possibility that emotion regulation may

involve memory does not seem unique to Gross and his colleagues' model; rather it seems characteristic of the fields of emotion regulation and memory in general (for similar observations, see del Palacio-Gonzalez, Berntsen, & Watson, 2017; Holland & Kensinger, 2010). Hence, it appears worthwhile to develop a comprehensive theoretical framework that elucidates the relationship between the two phenomena.

The relationship between emotion regulation and memory may not just benefit from theoretical specification but also from empirical exploration, although some data has already been obtained. Some research has found that diverse regulation strategies may have different effects on subsequent memory. For example, the cognitive change strategy of reappraisal (e.g. reinterpreting a threatening stimulus) leaves intact or improves subsequent memory (probably because of elaborate processing), but the response modulation strategy of expressive suppression (e.g. putting on a calm face while experiencing anxiety) tends to degrade memory (likely because of mental load; e.g. Ahn et al., 2015; Richards & Gross, 2000). However, such research has investigated how regulation affects memory encoding and relatively little research has examined how it may involve existing memories. One study found that peoples preferences for recalling different emotional (e.g. exciting or fearful) memories depended on the goals (e.g. approach or avoidance) of subsequent tasks (Tamir & Ford, 2009). Another study found that the attentional deployment strategy of focusing away from (rather than towards) the emotional content of memories decreased the intensity of emotional experience (Denkova, Dolcos, & Dolcos, 2015). Moreover, three studies found that the cognitive change strategy of reappraisal may downregulate emotional experiences stemming from recalling negative memories (Kross, Davidson, Weber, & Ochsner, 2009; Fabiansson, Denson, Moulds, Grisham, & Schira, 2012; Holland & Kensinger, 2013) and one of them reported prolonged (half an hour) effects on the intensity of memories (Holland & Kensinger, 2013). Although limited research has explicitly investigated the relationship between emotion regulation and memory, research from other domains can be interpreted as pertaining to this relationship, as will be argued in the following pages.

Mnemonic emotion regulation

I propose that a focus on retroactive regulation is important for understanding not only immediate but also enduring effects of emotion regulation and that

the term mnemonic emotion regulation may be used whenever attempts are made to control memories to regulate emotional experiences and expressions. Mnemonic emotion regulation may encompass quantitative control of memory accessibility. This may involve momentarily increasing access to a memory and/or making it more likely that it will be retrieved in the future and possibly in many different situations, for example by rehearsing a pleasant memory. This process has been labelled positive control (Levy & Anderson, 2008), facilitation (Conway, 2009) and the up-tendency (Erdelyi, 2010) and it may result in memory excitation and remembering. Alternatively, quantitative regulation may involve the opposite process, namely momentarily decreasing access to a memory and/or making retrieval less frequent over time, for example by attempting to continuously avoid an unpleasant memory. This process has been termed negative control (Levy & Anderson, 2008), suppression (Anderson & Green, 2001) and the down-tendency (Erdelyi, 2010) and it may result in memory inhibition and forgetting. Mnemonic emotion regulation may also encompass qualitative control of memory characteristics. This may involve altering or creating the content of a memory, for example by trying to replace negative with positive details of a memory. Alternatively, it may involve changing the context or interpretation of a memory, such as when linking a negative and a positive memory. This process has been called reconstruction (Bartlett, 1932/1964), distortion (Shane & Peterson, 2004) and elaboration (Erdelyi, 2006) and it may result in memory modification and construction. Overall, increasing access to, decreasing access to and altering characteristics of memories may be three distinct mnemonic emotion regulation processes (see Figure 1).

Mnemonic emotion regulation may be achieved by a working self that is part of working memory and comprises a conceptual self (e.g. self-beliefs and values) and a goal system (e.g. plans and projects) which processes memories (see Conway, 2009; Conway et al., 2004), contingent on executive functions (e.g. on the ability to sustain attention and to inhibit inappropriate responses; see Levy & Anderson, 2008). The working self may use emotional memories as a basis for emotion regulation (e.g. retrieve a positive memory to elicit joy) or target memories to change them (e.g. attempt to permanently increase access to a positive memory). Also, the working self may mostly be concerned with regulating autobiographical and episodic (spatio-temporally contextualised) memories. Such memories are self-related and often involve vivid sensory images why they seem especially likely to evoke emotions and be the focus of regulative efforts. For example, when preparing a presentation, an individual may practice reliving memories of previous successes to elicit positive emotions that confirm and energise a belief in the self's capabilities. Semantic (factual and general) memories may be less obviously related to emotion regulation because they are decontextualised and abstract. However, retrieving concepts that have emotional connotations might be a basis for regulation, such as when focusing on concepts like "determination" and "strength" creates a positive frame of mind. Alternatively, retrieving factual and schematic knowledge may be a basis for regulation, such as when remembering the fact that airplanes almost never crash and using a generic memory to imagine being on a sunny beach helps reduce fear of flying and even makes a flight a pleasant experience. In the following pages, I will focus on emotion regulation of autobiographical and

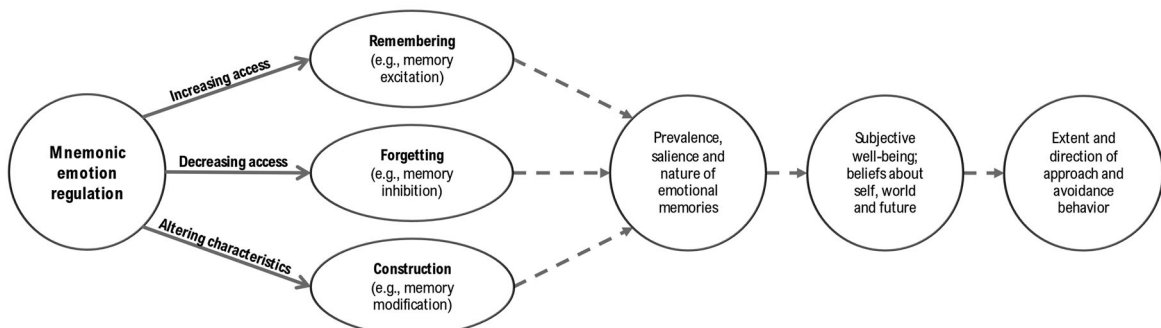


Figure 1. Mnemonic emotion regulation. Mnemonic emotion regulation may increase or decrease access to or alter the characteristics of memories. Such regulation may influence the prevalence, salience and nature of emotional memories, which may affect subjective well-being and important beliefs as well as approach and avoidance behaviour. Please note that the dotted lines are meant to illustrate more tentative relations.

episodic memories because it is probably the most prevalent type of regulation.

Mnemonic emotion regulation may be part of the three functions that autobiographical remembering has traditionally been suggested to serve, namely self-, directive-, and social functions (Pillemer, 1992; also see Pasupathi, 2003). Such regulation may protect, enhance and/or ensure the consistency of the self (Skowronski, 2011) and thereby enable coherence and continuity in identity. For example, the self may be protected when self-threatening feedback is forgotten (e.g. Sedikides & Green, 2000). Also, mnemonic emotion regulation may direct an individual when mostly positive memories are retrieved, and negative memories are made less accessible. Positive rather than negative memories are associated with high self-efficacy (Bandura, 1977) and typically concern goal-attainment and success (Singer & Salovey, 1993). Accordingly, such memories are well-suited as a basis for prospective simulations that may help productive problem-solving and future planning (see Schacter, Addis, & Buckner, 2007). For example, remembering a well-done presentation may serve as a fruitful foundation for preparing an upcoming performance. Finally, such regulation may serve social functions. Retrieving mostly positive rather than negative memories is likely to prime a good mood which is socially desirable, and which may facilitate interaction and bonding. Additionally, sharing memories of successes rather than failures may lead to believing in an individual's capabilities and evoke appreciation and respect. As alluded to here, people may often want to hold on to positive memories that are pleasant and encouraging and avoid negative memories that are unpleasant and counterproductive and thereby engage in simple hedonic regulation. However, sometimes mnemonic emotion regulation may be instrumental. I consider both possibilities in the following pages.

Increasing access

People may selectively increase access to certain memories because of hedonic motives and thereby engage in quantitative mnemonic emotion regulation. Selectively retrieving positive memories might be a basis for lifting a neutral state or boosting an already pleasant one. For instance, an individual may focus on recalling a compliment delivered by a colleague to feel pride and worth. Yet, selective retrieval of positive memories may also counter unpleasant affect.

Retrieving such memories may dampen negative emotions and stress (Speer & Delgado, 2017) and lead to mood-repair (Josephson, Singer, & Salovey, 1996). Healthy individuals can use recall of positive memories to improve a negative mood (e.g. Joormann, Siemer, & Gotlib, 2007; Josephson et al., 1996; Seebauer et al., 2016). For example, Joormann and her colleagues had participants watch a sad film to induce a negative mood. Then approximately half of them retrieved positive memories from their high school years while the other half engaged in a distraction task. Both tasks improved the participants' mood to a comparable degree, thereby demonstrating that selective retrieval of positive memories may boost mood. However, some factors moderate the efficacy of mood-incongruent recall. One factor is the mode of processing; engaging in concrete (detail-focussed and imaginative) rather than abstract (analytical and explanatory) processing of memories leads to mood-repair (Seebauer et al., 2016; Werner-Seidler & Moulds, 2012), probably because the former but not the latter processing mode involves retrieving sensory details and images, which may be especially effective in evoking emotions. Another factor that possibly moderates the effectiveness of mood-incongruent recall is memory content. Seebauer and her colleagues (2016, Experiment 1) found that retrieving social rather than achievement memories most efficiently led to mood-repair, although they did not replicate this finding subsequently (Experiment 2).

People may selectively retrieve and use positive memories as a basis for improving the current mood. However, they may also target such memories to increase their salience and prevalence over time. In fact, two recent studies have shown that the application of certain mnemonic methods may increase long-term retention of positive memories (Dalglish et al., 2013; Werner-Seidler & Dalglish, 2016). For example, Werner-Seidler and Dalglish deployed the classic method of loci (MoL) with healthy participants (who had recovered from depression) and investigated if the use of it would facilitate mood-repair and long-term retention of positive memories. Participants generated 15 self-affirmative personal memories and were then assigned to either a MoL group in which they were to associate the memories with a familiar route using bizarre images or a rehearsal group in which they were to use repetition or any other helpful strategy to retrieve the memories. Then, participants practiced recalling the positive memories for approximately 5 weeks using either

method (three times during the first week and one time during the following weeks). Surprise recall tests administered one week and three months later showed that participants in the MoL group remembered more of the memories than participants in the rehearsal group. Also, participants in the MoL group reported using positive memory retrieval as a mood-repair strategy more often than participants in the rehearsal group (from one to five weeks after the beginning of the study). Furthermore, participants in both groups were more successful in their ability to improve positive affect by retrieving pre-chosen positive memories after a negative mood induction procedure five weeks later compared with a third group of participants who had not practiced retrieval. Such research demonstrates how deliberately practicing recall may improve long-term retention of positive memories.

More generally, experimental research has identified several methods that may aid long-term retention, and which may potentially be used to increase long-term access to positive memories (see Dunlosky, Rawson, Marsh, Nathan, & Willingham, 2013). First, retrieval per se (e.g. selectively retrieving positive memories, as described earlier) may facilitate long-term retention. Research on practice testing shows that continuously being prompted to recall some information improves its accessibility (Roediger, Putnam, & Smith, 2011). It is possible that continuously testing the content of appreciated positive memories will increase their state of activation over time. This idea aligns with research which shows that people who deliberately recall a few good things that happened to them every day for a period, report an increase in happiness (Seligman, Steen, Park, & Peterson, 2005), probably because this exercise makes positive memories salient. Second, elaboration during encoding or rehearsal improves long-term retention (Craik, 2002). Elaboration may involve using the MoL, as depicted previously, but it may also involve thinking about the implications of such experiences and integrating them with existing knowledge to facilitate long-term retention. For example, people might reflect on and attempt to incorporate positive experiences with desirable aspects of the self and their life scripts (e.g. reminiscing about having made many new friends during the twenties may confirm the self as prosocial and also be age-expectable behaviour; see Rubin & Berntsen, 2003). Such processing may lead to positive memories that are highly complex and inter-associated, and which may therefore be accessed

with great speed and efficacy (Unkelbach, Fiedler, Bayer, Stegmüller, & Danner, 2008). Third, retrieval of positive memories may be facilitated by appropriate cues in the environment (see Tulving, 1974). Arranging and seeking out reminders of positive experiences may help an individual to often recall such experiences. For example, uploading pictures of enjoyable experiences on a Facebook wall may cue an individual to often remember such experiences.

In accordance with the proposition that healthy people can and want to focus on positive memories for hedonic reasons, several studies have found that healthy people tend to mostly retrieve such memories (e.g. Bohn & Berntsen, 2007; Breslin & Safer, 2011; Walker, Skowronski, Gibbons, Vogl, & Ritchie, 2009). For example, in the study by Walker and his colleagues (2009) participants retrieved autobiographical memories and retrospectively estimated how frequently different types of memories were recalled and for what reason. According to the participants, they most often recalled memories for social communication, and then to re-experience an emotion, to maintain memory details, involuntary (e.g. prompted by cues in the environment), and because of a desire to understand an event, respectively. In all instances of intentional retrieval, but not with regards to involuntary retrieval, positive rather than negative memories were most often recalled. However, even regarding involuntary memories, it seems as if most other studies have found that they are primarily positive (see Rasmussen, Ramsgaard, & Berntsen, 2015), possibly because people seek out likable environments that contain many positive reminders. Also, people who are part of a group (e.g. fans of a baseball team) often rehearse more positive than negative memories and this may lead the former rather than the latter memories to become more easily accessible and vivid (Breslin & Safer, 2011; Pennebaker & Gonzales, 2009). Overall, positive rather than negative memories are prioritised and prevalent among healthy people; they may be desirable to retrieve regularly because they are enjoyable and pleasurable in themselves but also because they serve self-, directive-, and social functions. Such memories may affirm the self, indicate promising courses of action and facilitate relationships.

Decreasing access

People may selectively decrease access to certain memories because of hedonic motives and thereby

engage in another type of quantitative mnemonic emotion regulation. For example, a reminder may trigger a memory of an unpleasant encounter and the memory may be attempted avoided because it evokes anger. Research on memory suppression shows that people may intentionally avoid certain memories, as in the think/no-think (TNT) paradigm. In this paradigm, participants train cue-response associations. Then, they are continuously presented with cues and suppress (avoid thinking about) some responses while rehearsing (practice thinking about) other responses. A final test often shows that cued recall of to-be-suppressed memories is less than cued recall of memories that were trained but were not prompted in the critical phase of the experiment (e.g. Anderson & Green, 2001; also see Anderson & Hanslmayr, 2014). Studies have shown that memories of negative words (e.g. Van Schie, Geraerts, & Anderson, 2013), pictures (e.g. Küpper, Benoit, Dalgleish, & Anderson, 2014) and autobiographical experiences (e.g. Noreen & MacLeod, 2014) can be suppressed. For example, Noreen and MacLeod had healthy participants generate positive and negative autobiographical memories in response to neutral cues. Then, participants practiced not thinking about memories in response to some cues while practicing thinking about memories in response to other cues. Finally, in a test of cued recall, participants exhibited degraded recall of both types of memories. Below-baseline forgetting of negative memories in the TNT paradigm suggests that people can purge such memories from awareness and may be evidence of an inhibitory control mechanism that can decrease access to unwanted memories (Anderson & Hanslmayr, 2014; but see Raaijmakers & Jakab, 2013 and note that suppression of negative memories is not always found, e.g. may fail with consolidated aversive memories, Liu et al., 2016). Memory suppression may help people avoid being automatically carried away whenever they are faced with reminders of negative experiences.

People may also forget negative experiences as part of updating their memories. For example, initial failures to solve a recurrent task at work may be forgotten and updated with knowledge about subsequent successes. This type of forgetting is mimicked by experiments with list-method directed forgetting. In such experiments, people learn a list of items (List 1; e.g. words, pictures or autobiographical memories) and are then led to believe that it should be forgotten and that another list of items (List 2)

should be learned instead. Several studies have found that an instruction to forget negative list 1 items can decrease performance on a subsequent unexpected test of recall (e.g. Barnier et al., 2007; Patrick & Christensen, 2013; Wessel & Merckelbach, 2006). For example, Barnier and her colleagues (2007) reported a similar magnitude of directed forgetting of negative, positive and neutral self-provided autobiographical memories (cued by thematic words such as “hate”, “love” and “supermarket”) across four experiments in free but not in cued recall. Such studies show that negative memories may be forgotten as part of updating memory. Directed forgetting may permit updating of negative memories with neutral or positive information and has been explained by suppression and resulting memory inhibition (Bjork, Bjork, & MacLeod, 2006; but see Sahakyan & Kelley, 2002 and note that list-method directed forgetting of negative memories sometimes fails, e.g. in high-arousal contexts, Minnema & Knowlton, 2008). Overall, people may often be able to forget and replace old negative memories with new neutral or positive memories, which may make them less worried about and not too distracted by past failures and struggles.

The importance of suppression for long-term memory seems at least partly dependent on whether it can lead to durable memory inhibition that lasts for more than a few hours. Some TNT studies have found no evidence of inhibition when testing memory for words after a delay of one week (Meier, König, Parak, & Henke, 2011; Nørby, Lange, & Larsen, 2010) or memory for self-relevant episodic information after 3-4 months (Noreen & MacLeod, 2014, Experiment 2) or 12-13 months (Noreen & Macleod, 2014, Experiment 1). These findings show that suppression does not necessarily have long-term effects. However, other results indicate that suppression may affect memory for more than a short time. One TNT study found inhibition to still be present after 24 hours (Hotta & Kawaguchi, 2009). Also, Noreen and MacLeod (2014) showed that participants who were most successful in forgetting to-be-suppressed memories in the immediate test were also least likely to exhibit rebound effects in the delayed tests. Furthermore, one recent study demonstrated delayed effects of suppression of negative memories (Streb, Mecklinger, Anderson, Lass-Henne-mann, & Michael, 2016). In this study, participants watched an aversive movie and individuals with high suppression ability as measured via their

performance in the TNT paradigm had fewer and less distressing memory intrusions in the following days, compared with individuals with low suppression ability. Finally, one study which did not use the TNT, but a related paradigm, found that people who were skilled at forgetting recalled fewer negative autobiographical memories than participants who were poor at forgetting (Storm & Jobe, 2012; also see Giebl, Storm, Buchli, Bjork, & Bjork, 2016). Results from the latter studies indicate that suppression can decrease access to negative memories for at least some days. Accordingly, individuals with efficient executive functions and/or who persistently utilise suppression might succeed in keeping everyday unwanted memories and associated emotions out of consciousness enduringly.

Other methods than suppression can decrease access to memories. Forgetting can be caused by interference; a memory may become inaccessible because other information competes with and blocks access to it. People may intentionally instigate interference (e.g. Del Prete, Hanczakowski, Bajo, & Mazzoni, 2015; Hertel & McDaniel, 2010; Joormann, LeMoult, Hertel, & Gotlib, 2009). For example, Joormann and her colleagues reported that healthy participants could use positive substitutes to avoid thinking about and forget negative responses in the TNT paradigm. Retrieving positive memories to distract oneself from negative reminiscences may be a productive mnemonic emotion regulation strategy and it seems to rely on a different neural network than memory suppression (Benoit & Anderson, 2012). That interference can cause forgetting of negative memories is also evidenced by research on reconsolidation. Traditionally, consolidated memories were believed to be stable, but recent research has shown that retrieval may sometimes (e.g. when it involves a prediction error regarding a subsequent event) bring an old unpleasant memory into a labile state where it can be interfered with and influenced, for example when old information is replaced with new information (e.g. Schiller et al., 2010; also see Lee, Nader, & Schiller, 2017). Another method that may decrease access to negative memories or at least make such memories less intense is expressive behaviour. Expressive writing may lower the negative affect associated with aversive events (Pennebaker, 1997) although the effects on long-term memory is unclear (Rubin, Boals, & Klein, 2010). Also, sharing negative memories with other people may help dissipate unpleasant affect, especially over time (Walker et al., 2009). A further method that may decrease

access to negative memories is to avoid reminders, thereby generating cue-dependent forgetting (Tulving, 1974). When certain cues are avoided, it is less likely that corresponding unwanted memory representations come to mind.

Altering characteristics

People may alter the characteristics of their memories and thereby engage in qualitative mnemonic emotion regulation, for hedonic reasons. For example, initial negative impressions of a new colleague may be reconstructed to facilitate pleasant and productive interactions. The view that memories can be reconstructed based on post-event information and current attitudes and goals is prevalent (e.g. Bartlett, 1932/1964; Greenwald, 1980; Ross, 1989) and some consider all autobiographical memories to be constructed and false to some degree (Conway & Loveday, 2015). Memory construction for the purpose of emotion regulation may involve altering the content of a memory. Neisser (1981) famously showed that John Dean's memory of his conversations with President Nixon during the Watergate crisis involved self-serving distortions. Other studies have shown that people may positively distort their recall of their educational achievement (Bahrack, Hall, & Berger, 1996; Bahrack, Hall, & Da Costa, 2008). For example, Bahrack and his colleagues (2008) found that people accurately recalled most of their college grades from 1 to 54 years earlier, but also that they produced several commission errors. Most of these errors inflated the grade and they occurred especially often among the best students regarding the courses they enjoyed most. The results are explainable by the supplementary theory of autobiographical memory distortion (Bahrack et al., 2008) which holds that retrieval involves both reproductive and reconstructive processes (also see Reyna & Brainerd, 1995). Specifically, when reproductive retrieval fails, reconstructive processes supposedly fill memory gaps based on generic knowledge (e.g. about what is typical) and hedonic motives (e.g. to minimise negative emotion). Systematic distortions are seen as most likely to target negative memories that are seldom rehearsed and may be modified to experience more gratifying emotions. Memories from other domains than the educational (e.g. the social domain) might possibly become positively biased as well. However, studies on self-induced changes of emotional memories seem scarce, probably due to methodological challenges.

Mnemonic emotion regulation may not only involve changing the declarative and explicit but also the emotional and visceral content of memories. Often, the correlation between the type and intensity of event-elicited emotions and retrospective ratings of such emotions is high, but that is not always the case (see Levine, Lench, & Safer, 2009). For example, a failed job application, which was originally associated with sadness, may be recalled with relief when finding out that the work place is very stressful. More generally, research has demonstrated that post-event information processing may influence remembered emotions. In one study, participants reappraised negative memories and lowered the intensity of the associated emotions, with the effect persisting for at least half an hour (Holland & Kensinger, 2013). In another study, students recalled how anxious they had felt up to an exam, either before or after receiving the grade (Safer, Levine, & Drapalski, 2002). Compared to students who had not yet been told their grades, students who did well underestimated their pre-exam anxiety and students who did poorly overestimated this anxiety. Two other studies found that people's current attitudes towards US presidential candidates biased their recall of the emotions they felt during the campaigns and the elections (Levine, 1997; Wilson, Meyers, & Gilbert, 2003). Also, a further study found that the intensity of grief people remembered experiencing six months after the death of a spouse was more highly correlated with the current level of grief measured after five years than the actual level of grief reported after six months (Safer, Bonanno, & Field, 2001). Not only the intensity but also the valence associated with an event may change over time. Walker and Skowronski (2009) describe the phenomenon of flexible affect, namely that the emotional valence may occasionally switch from event occurrence to recall, mostly from negative to positive. Post-event appraisals, involving current attitudes, emotions and coping efforts, may explain the change of memory for emotions over time (see Walker & Skowronski, 2009). Overall, this research demonstrates that people may reappraise past events and thereby regulate the emotional content of associated memories.

People may also construct false memories of events that never happened for emotion regulation purposes. In such instances mental content is entirely fabricated, but nonetheless experienced as memory. Constructing a false positive memory of an entire event might help people experience positive

emotions. For instance, falsely remembering having been praised by a teacher for doing a presentation in school many years ago might lead an individual to feel joy and pride. In support of the idea that false positive memories may be constructed, some studies have shown that people can be led to remember that they have had a positive experience with food or drink (e.g. Laney, Morris, Bernstein, Wakefield, & Loftus, 2008; Laney, Kaasa et al., 2008; Mantonakis, Wudarzewski, Bernstein, Clifasefi, & Loftus, 2013). For example, Laney, Morris and their colleagues misinformed adult participants that they had enjoyed asparagus as children (e.g. "you loved asparagus the first time you ate it"). Several participants came to believe or remember that they liked this food as children, thereby showing that false positive beliefs and memories can be implanted. Other misinformation studies have demonstrated that adults can be persuaded that they have experienced entire fictitious events which are exciting albeit not necessarily unequivocally positive, for example shaking hands with Bugs Bunny (Braun, Ellis, & Loftus, 2002) or taking a hot air balloon ride (Wade, Garry, Read, & Lindsay, 2002) in childhood. Accordingly, false positive memories of entire events may be constructed, probably with the help of elaborative and imaginative processing. Such false memories may evoke positive emotions and sometimes also have beneficial behavioural consequences, for example a desire to eat healthy food (Laney, Morris et al., 2008). However, there seems to exist few positive compared to neutral or negative memory implantation studies. Also, not much seems to be known about spontaneously occurring false positive memories.

Not only the content but also the context of memories may be changed or constructed for emotion regulation purposes. A memory may be re-associated with other memories. For example, a memory of a failure that was originally tied to other memories of defeat may be re-associated with memories of struggles that led to growth, thereby providing a context of hope. In general, a memory may be thought about differently to change its affective tone. For example, Ross (1989) proposed that people reconstruct memories based on current appraisals in combination with implicit theories about whether these have changed over time. If an individual believes that appraisals have not changed he or she may be inclined to view the past as similar to the present, but if appraisals are believed to be

fluctuating, then the past may be constructed as better or worse than the present. In accordance with this idea, one study showed that people could distance themselves from past failures by extending subjective time (Ross & Wilson, 2002). In this study, participants reported feeling farther away from courses in which they had received a bad grade compared with courses in which they had received a good grade, when calendar time was controlled for. Other studies have experimentally manipulated the subjective time distance that people feel from bad and good events and shown that when people feel far away from (rather than closer to) a negative event they judge the self favourably (Wilson, 2000; Wilson & Ross, 2001). People may also distance themselves from disliked experiences by other contextual changes, for example by constructing a memory from a detached observer perspective rather than an intimate field perspective (e.g. Marchlewska & Cichocka, 2017; McIsaac & Eich, 2004; Vella & Moulds, 2014). Overall, such contextual changes may be considered instances of reappraisal (Ochsner & Gross, 2008) and allow for self-protection and experiencing fewer negative emotions or for self-enhancement and experiencing more positive emotions (see Newman & Lindsay, 2009).

Hedonic and instrumental regulation

On the surface, hedonic mnemonic emotion regulation may seem equivalent to focusing on positive memories and dismissing negative memories, as described previously. However, the relationship is certainly not that simple and some instances of what may be called atypical hedonic regulation can be identified. It may typically make people feel good when they focus on positive memories, but that is not always the case. If an individual is not satisfied with his or her current situation, retrieving distant positive memories may involve comparing a joyful past to a gloomy present and evoke negative emotions such as disappointment and sadness (see Joormann et al., 2007). The risk that unpleasant contrast effects may occur may sometimes make people inclined to avoid recalling certain positive memories. For example, pleasant memories associated with individuals or possibilities that have vanished may be avoided to prevent sadness and grief, such as when a loss leads people to suppress associated positive memories since they represent events that are unlikely to be (re-) experienced in the future (for a proposal regarding when

contrast effects occur, see Clark, Collins, & Henry, 1994). Moreover, although it may mostly make people feel bad when they focus on negative memories, retrieving such memories may sometimes give rise to positive emotions. When a negative past is interpreted in a self-improvement perspective in which an individual think back at distant negative experiences and note that the current situation is better, it may heighten the mood (Conway & Ross, 1984) and possibly evoke positive emotions such as relief and gratitude. Furthermore, socially sharing negative memories may help to dissipate associated unpleasant affect and evoke empathy (Rimé, 2007; Walker et al., 2009). Overall, hedonic regulation may occasionally involve dismissing positive memories and focusing on negative memories.

People regulate their emotional memories not only for hedonic but also for instrumental reasons; such memories may sometimes hinder or help goal pursuit. In some situations, it seems inappropriate to focus on and maybe even best to suppress memories of positive personal experiences, for example when trying to empathise with other people's misfortunes. Also, focusing on negative memories and related emotions may help obtain a personal goal in a certain situation, such as when remembering a dispute is associated with heightened arousal, combativeness and success in a confrontation (see Holland & Kensinger, 2013; Tamir & Ford, 2012). This might not only involve increasing access to negative memories but also distorting them, such as when a memory of an interaction with a foe is inflated to justify anger. Over time, focusing on negative memories may occasionally help establish identity and facilitate long-term goals. Dwelling on negative memories may be relevant to the current self, for example when recalling failures (e.g. failing to make people laugh) helps accepting and cementing a personal flaw (e.g. being an unfunny individual). Also, negative experiences (e.g. a rejection) are often informational regarding important adaptive goals (e.g. attachment) and remembering these may improve behaviour (e.g. analysing what caused a rejection may help finding an alternate way of approaching other people). Finally, it has been suggested that constructing entirely false memories of horrific events could help people explain their current difficulties and therefore to feel better (Fernández, 2015). However, although people may construct false memories of intensely negative events, it seems unlikely that constructing such memories may serve beneficial

emotion regulation purposes overall (Otgaar, Howe, Clark, Wang, & Merckelbach, 2015).

Discussion

Mnemonic emotion regulation may bias the memory system towards pleasant memories based on hedonic motives. According to this view, engaging the working self and exercising executive control may often involve promoting positive memories, dismissing negative memories and/or reconstructing unsatisfactory memories (see Anderson & Hanslmayr, 2014; Conway, 2009; Walker, Skowronski, & Thomson, 2003). Such efforts may in combination lead to a focus on mostly pleasant and productive memories in the present moment. However, the result may also be a more permanently skewed activation of memories so that mostly positive rather than negative memories are primed, salient and prevalent; this may be one reason why healthy people typically remember twice as many positive than negative memories (Walker et al., 2003; see Figure 1). A positively skewed memory system may allow for a positivity offset, that is, a mildly pleasant baseline activation of the information processing system (see Norris, Larsen, Crawford, & Cacioppo, 2011) and a high positivity ratio, that is, experiencing mostly positive rather than negative emotions over a given period (see Fredrickson, 2013). An abundance of positive emotions may be associated with at least two benefits. First, experiencing many positive emotions is associated with high ratings of subjective well-being, which is favourable because most people want to be happy (Kesebir & Diener, 2008) and because it may be tied to optimistic beliefs about the self, the world and the future. Second, a presence of mostly positive emotions may signal that things are going well and be associated with self-confident and explorative approach behaviour, which is favourable because it may build personal and social resources over time (Fredrickson, 2004). In accordance with this proposition, positive memories have been found to motivate productive behaviour, for example exercise (e.g. Biondolillo & Pillemer, 2015) and healthy eating (Laney, Morris et al., 2008).

Hedonic mnemonic emotion regulation may generally be beneficial in that it typically leads to high subjective well-being and approach behaviour. However, it may not always be so, for several reasons. First, if people are to align with reality and learn from failures they should probably only attempt to slightly bias but

not delude the memory system. Too much one-sided regulation may produce a past that is inauthentic and unrealistic (see Childress, 2015). For example, attempting to only remember positive experiences would seem extreme (see Fredrickson, 2013 who discusses drawbacks of excessive positivity) and it may be healthy to focus on negative experiences in situations that call for it (e.g. to focus on anger memories in confrontational situations; see Kim, Ford, Mauss, & Tamir, 2015). Second, although it may be beneficial to bias the memory system based on valence, competent regulation may also involve exercising control based on other mnemonic features. For example, negative experiences are useful to a different degree: Some such experiences are difficult to productively act on (e.g. feedback about a stable negative trait) and may best be forgotten but others are informative regarding how behaviour might be changed to avoid similar situations in the future and may best be remembered (e.g. feedback about a modifiable negative behaviour; see Green, Pinter, & Sedikides, 2005). Third, some people may benefit from not only occasionally but predominantly engaging in instrumental mnemonic emotion regulation due to enduring circumstances. For example, people in highly competitive and combative environments (e.g. professional sports) may continuously use fear and anger memories to sustain alertness and facilitate confrontations in order to progress (e.g. to win several boxing matches). It should be further explored which magnitudes and directions of mnemonic emotion regulation that best supports subjective well-being, mental health and adapting to different circumstances.

The process of regulation

The working self may dynamically control the accessibility, characteristics and organisation of autobiographical memory (see Conway, 2009; Conway et al., 2004) and it may determine whether hedonic mnemonic emotion regulation is prioritised. Simple hedonic regulation that involves promoting positive and dismissing negative memories and emotions may be most common, but a particular regulative effort ultimately appears contingent on present self-concepts, goals and circumstances and it may sometimes be atypical hedonic or counter-hedonic. However, regardless of the motives for a regulative effort, it may concern different aspects of a memory. First, such an effort may involve only parts of an episodic memory such as non-declarative episodic elements (implicit

somatomotor and visceral responses) rather than declarative interpretative frames (explicit conceptual knowledge structures; see Conway, 2009; Visser, Lau-Zhu, Henson, & Holmes, 2018). For example, research on reconsolidation has demonstrated that the bodily component of a fear memory (e.g. eyeblink startle) can be dampened or erased without affecting the declarative memory of the fear inducing incident (e.g. Schiller et al., 2010; also see Lee et al., 2017). Second, a regulative effort may concern different levels of mnemonic generalisation. It may involve episodic memories, but it might also involve memories on a higher level of abstraction, for example what has been termed general events and life-time periods (Conway, 2009; also see Berntsen, Rubin, & Siegler, 2011). Re-interpreting recurrent experiences and life-stories may change emotional experiences, such as when re-framing a lengthier crisis as educational rather than damaging elicits enthusiasm and hope instead of despair and resentment. Also, as noted previously, not only autobiographical and

episodic memories but also semantic memories might be involved in regulation.

The working self may instigate mnemonic emotion regulation at different time points in accordance with the process model (see Gross, 1998, 2015, and Figure 2). Approaching or avoiding reminders of positive and negative experiences may be part of the situation selection stage and influence the likelihood of eliciting emotional memories since retrieval is cue-dependent (Tulving, 1974). For instance, a physical context that is associated with an aversive experience may be avoided. Directing in situ mnemonic cues may be part of the situation modification stage, such as when trying to change the topic of a conversation to avoid prompts to recall distressing memories. Rehearsing and practicing retrieval or alternatively suppressing and interfering with memories may be part of the attentional deployment stage, which according to the model involves selective attention and distraction as means for emotion regulation (Ochsner & Gross, 2008). Altering the characteristics of a

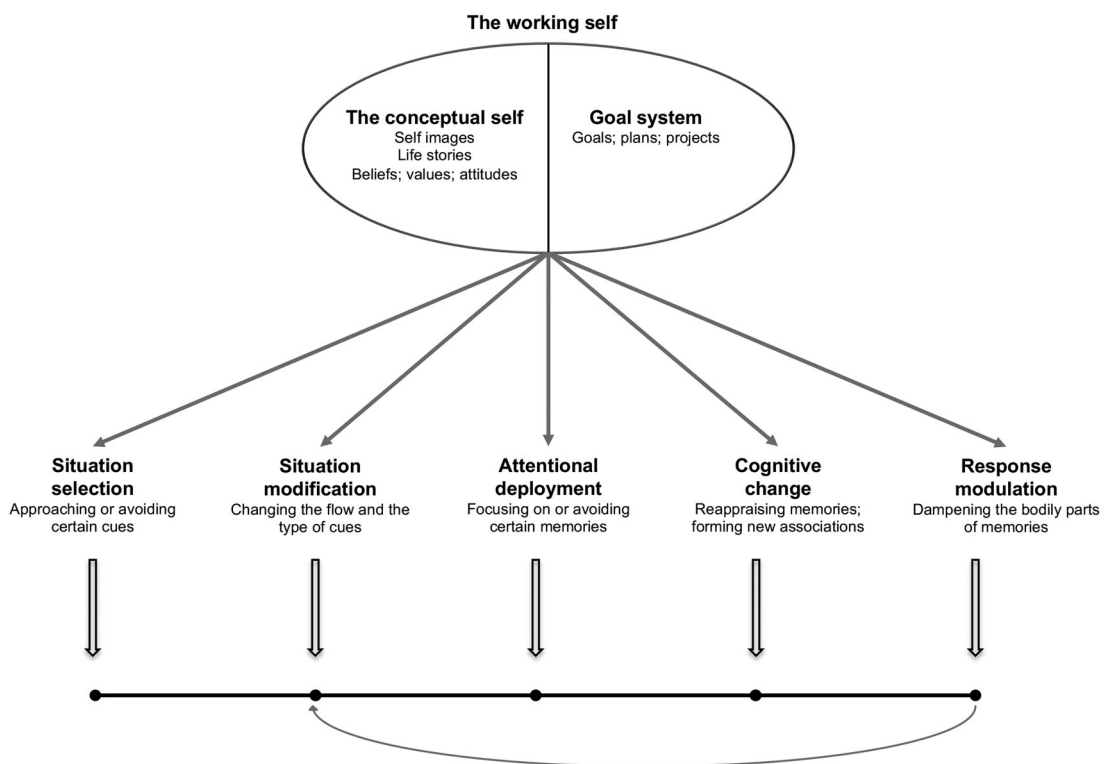


Figure 2. The mnemonic emotion regulation process. Mnemonic emotion regulation may target different stages in the emotion generative process as described by the process model (Gross, 2015). Also, it may be controlled by a working self (see Conway, 2009). As in Gross and his colleagues' original model, the backward arrow illustrates the dynamic nature of emotion regulation.

memory may be part of the cognitive change stage, which the model depicts as involving changing the meaning of an emotional stimulus. For example, reappraisal may change the content of a memory or involve relating a memory to a broad narrative, in accordance with the suggestion that regulation can involve abstract processing (see Lane, Ryan, Nadel, & Greenberg, 2015). Finally, trying to influence the expression of an emotional memory, such as when using a relaxation technique to dampen its bodily manifestation, may be part of the response modulation stage. This perspective on the process model invites considering emotion regulation strategies that uniquely involve using memory as well as the long-term (learning) effects resulting from utilising strategies that target memory. Using and targeting a memory may be similar processes that both change it; the former involves retrieval which can be a memory modifier (Bjork, 1975; also see Roediger et al., 2011) and the latter involves influencing long-term retention. Alternatively, these processes may be somewhat different because the primary goal of using a memory is to regulate current emotions, possibly without an explicit intention to change the memory, while the primary goal of targeting a memory by definition is to change it to regulate recurrent emotions, perhaps based on metamnemonic considerations. It seems worthwhile to further explore which mnemonic emotion regulation strategies are most efficient, have enduring effects on emotions and to what degree using and targeting memories are similar processes.

Individual differences in regulation

Individual factors are likely to influence mnemonic emotion regulation. Individuals might be characterised by different regulation styles. For example, some people (e.g. those who employ a repressive coping style; see Alston et al., 2013) may mostly engage in one mnemonic emotion regulation process (e.g. attempts to decrease access to negative memories) but not others (e.g. attempts to increase access to positive memories). Other people may engage in all the three regulation processes that have been depicted earlier. Moreover, individuals who can be characterised by broadly similar regulation styles may differ with respect to which specific methods they use. For instance, one individual may mostly engage in suppression while another individual may prefer to retrieve positive substitutes to

decrease access to a negative memory. Furthermore, individuals may not only differ in their preferences but also their capabilities for mnemonic emotion regulation. Some people may be highly effective at controlling their memories, for example due to potent executive functions (Levy & Anderson, 2008). Other people, such as those who suffer from affective disorders, may have difficulties with control. For example, depression is associated with problems forgetting negative memories (e.g. via direct suppression; Joormann et al., 2009) and feeling positive memories (e.g. via concrete processing; Werner-Seidler & Moulds, 2012; mnemonic control may also be moderated by the context, such as when control is difficult because the environment affords reminders of unwanted memories; see Hirst & Echterhoff, 2012). Regulation styles may to some extent be acquired in infancy. For example, parents who reminisce together with their children in highly elaborative ways (e.g. by interpreting aversive experiences in detail) may help teach them to regulate emotional experiences particularly well (Fivush & Sales, 2006). Future research should develop a taxonomy of mnemonic emotion regulation styles, investigate traits that make regulation likely to succeed and trace the development of regulation.

If individuals differ in their ability to deploy mnemonic emotion regulation and if effective regulation is associated with favourable outcomes then it may sometimes be useful to improve it (see Dalgleish & Werner-Seidler, 2014; Visser et al., 2018). One possibility is to train executive functions that underlie mnemonic emotion regulation, for example by training emotional working memory (Schweizer, Grahn, Hampshire, Mobbs, & Dalgleish, 2013). Another possibility is to acquire more efficient memory control strategies. The MoL may heighten the salience and prevalence of positive memories, but other methods also seem usable. For example, memory specificity training may involve learning to elaborate on positive memories and to evoke sensory details to remember and not just think about but also feel the memories (see Raes, Williams, & Hermans, 2009). It may also be possible to train the ability to downregulate access to memories, for example by improving the capability to use suppression (Anderson & Hanslmayr, 2014) or distracting substitutes (Joormann et al., 2009) to forget unwanted memories. Also, training may involve learning to use retrieval to induce memory plasticity, thereby making it possible to weaken and reconsolidate memories, by behavioural (e.g. via

extinction training) or pharmacological (e.g. administration of a beta-adrenergic receptor blocker) means (see Lee et al., 2017). Furthermore, adopting an observer rather than a field perspective when recalling a memory generally appears to decrease the intensity of associated emotions (Wallace-Hadrill & Kamboj, 2016). Finally, although the capability to alter or fabricate mnemonic content may be boosted, for example by improving upon the ability to employ misinforming (self-) suggestions to (re-) construct memories, it is reality-distorting (Childress, 2015; but see Sher, 2000). However, using re-appraisal to productively reframe negative experiences does not necessarily alter the content and authenticity of memories and seems integral to much psychotherapy. The efficacy of different training regimes should be further explored in an ethically responsible way.

Final thoughts

To sum up, I have proposed that it is important to not only focus on proactive and reactive but also retroactive emotion regulation that involves memory. Mnemonic emotion regulation may help people control their emotional past, so they are not reflexively carried away by it. People may engage the working self and executive functions to increase or decrease access to or alter the characteristics of their memories to influence the experience and expression of emotions. Mnemonic emotion regulation may mostly be hedonically motivated, bias the memory system towards positive memories and serve self-, directive- and social functions. It may help people focus on pleasant and productive memories on a moment-to-moment basis. However, mnemonic emotion regulation is unique in that it may often have long-term consequences. Hence, such regulation may lead to a more permanent abundance of positive memories and emotions, which seems favourable because it provides people with a mostly pleasant setpoint and lead them to be hopeful and engage with the world. On the other hand, mnemonic emotion regulation is not always conceivable since some individual traits may make it less likely to succeed (e.g. ruminative tendencies) and because some contexts may constrain memory control (e.g. contain salient reminders of unwanted experiences). Also, mnemonic emotion regulation may not only be engaged for hedonic reasons; it may be engaged for instrumental or personal idiosyncratic reasons as well. The reasons for and the functionality

of regulation in a particular instance ultimately depend on present self-concepts, goals and circumstances. Overall, by attempting to control the access to or the characteristics of certain memories it may be possible to reform the past and build a mnemonic foundation for a desired future.

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